



**SIMATS ENGINEERING
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SCIENCES**



AI-Based Sustainable Agriculture Management System

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ABSTRACT



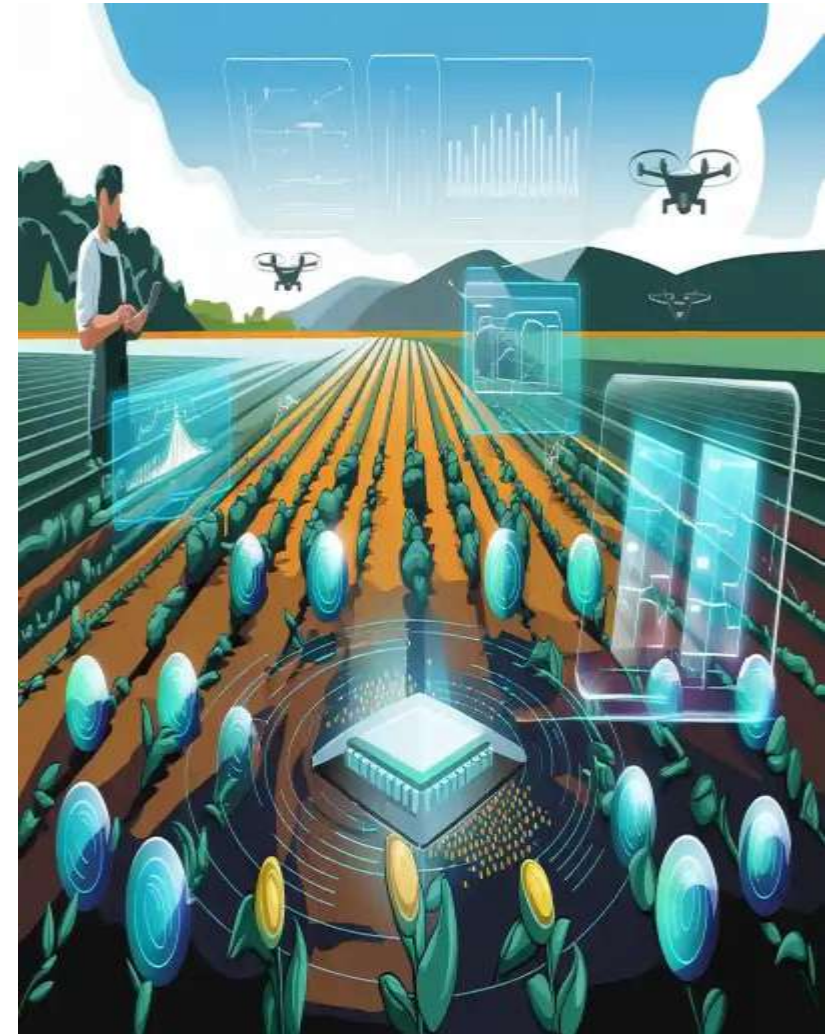
- Develops an AI-based sustainable agriculture management system using Java to improve farming efficiency and productivity.
- Analyses soil, weather, and crop data to recommend the most suitable crops for cultivation.
- Predicts crop diseases and pest attacks at an early stage using Artificial Intelligence, helping reduce crop losses.
- Promotes sustainable farming by increasing crop yield, reducing resource wastage, and supporting environmentally friendly agricultural practices.
- Uses AI-driven decision support to help farmers make accurate and timely farming decisions.
- Maintains farmer, crop, and field information in a secure database for efficient farm management and record keeping.
- Improves agricultural productivity while reducing operational costs, contributing to food security and sustainable resource management.



INTRODUCTION

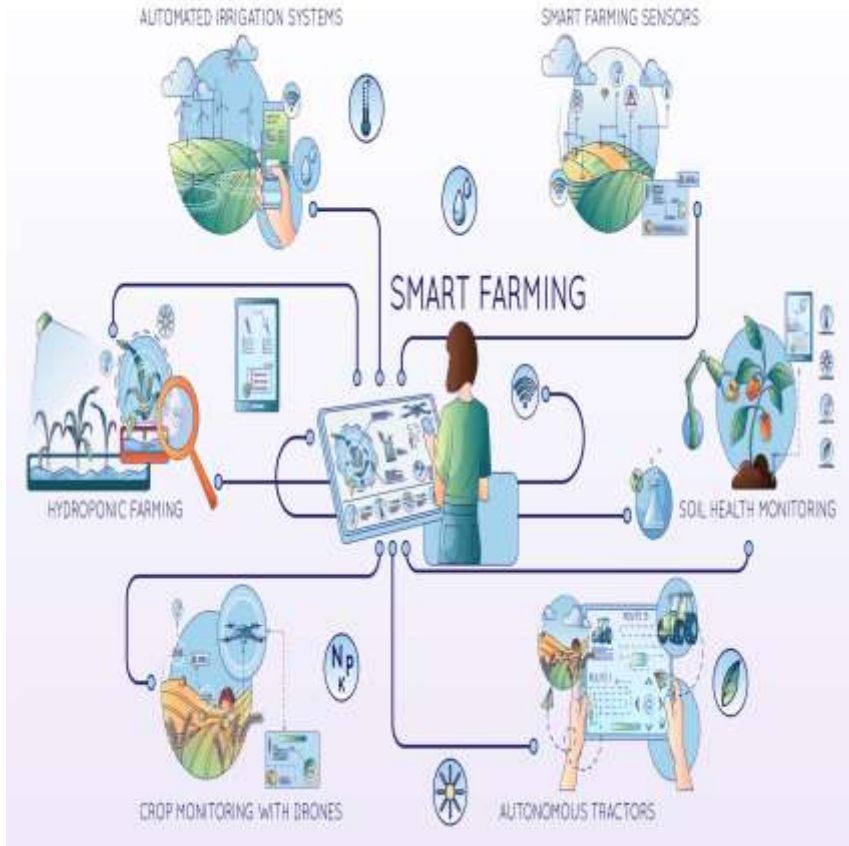


- Agriculture is one of the most important sectors, providing food, employment, and raw materials for economic development.
- Traditional farming methods often face challenges such as unpredictable weather, poor soil management, water scarcity, and crop diseases.
- Artificial Intelligence (AI) helps farmers make better decisions by analyzing agricultural data and providing accurate recommendations.
- The proposed system is developed using Java to create a reliable, user-friendly, and efficient agriculture management application.
- The system analyzes soil, weather, and crop data to recommend suitable crops and improve farming practices.





OBJECTIVES



- To develop an AI-based agriculture management system using Java for efficient and smart farming.
- To analyze soil, weather, and crop data and recommend the most suitable crops for cultivation.
- To predict crop diseases and pest attacks at an early stage using Artificial Intelligence.
- To optimize irrigation and fertilizer usage for better resource management and reduced wastage.
- To improve crop productivity while promoting sustainable and eco-friendly farming practices.



LITERATURE SURVEY



Author(s)	Year	Approach / Models	Key Idea	Limitations
Pervez, A. K. M., Kabir, M. S., et al.	2026	AI Trends in Sustainable Agriculture	Reviews recent AI techniques for sustainable farming and efficient resource utilization.	Mostly theoretical; lacks implementation of scheduling algorithms.
Sabharwal, A., & Reddy, S. R. N.	2026	AI + IoT for Smart Farming	Integrates AI and IoT for soil nutrient management and intelligent irrigation systems.	Higher implementation cost and infrastructure requirements.
Ali, Z., Muhammad, A., Lee, N., et al.	2025	AI-Driven Smart Agriculture	Artificial Intelligence supports crop monitoring, disease prediction, and intelligent decision-making.	Requires large and high-quality datasets for accurate predictions.
Rajwar, K., Pant, M., & Deep, K.	2024	Machine Learning in Agriculture	Machine learning techniques improve crop prediction, resource management, and farming efficiency.	Limited real-time scheduling and task prioritization.

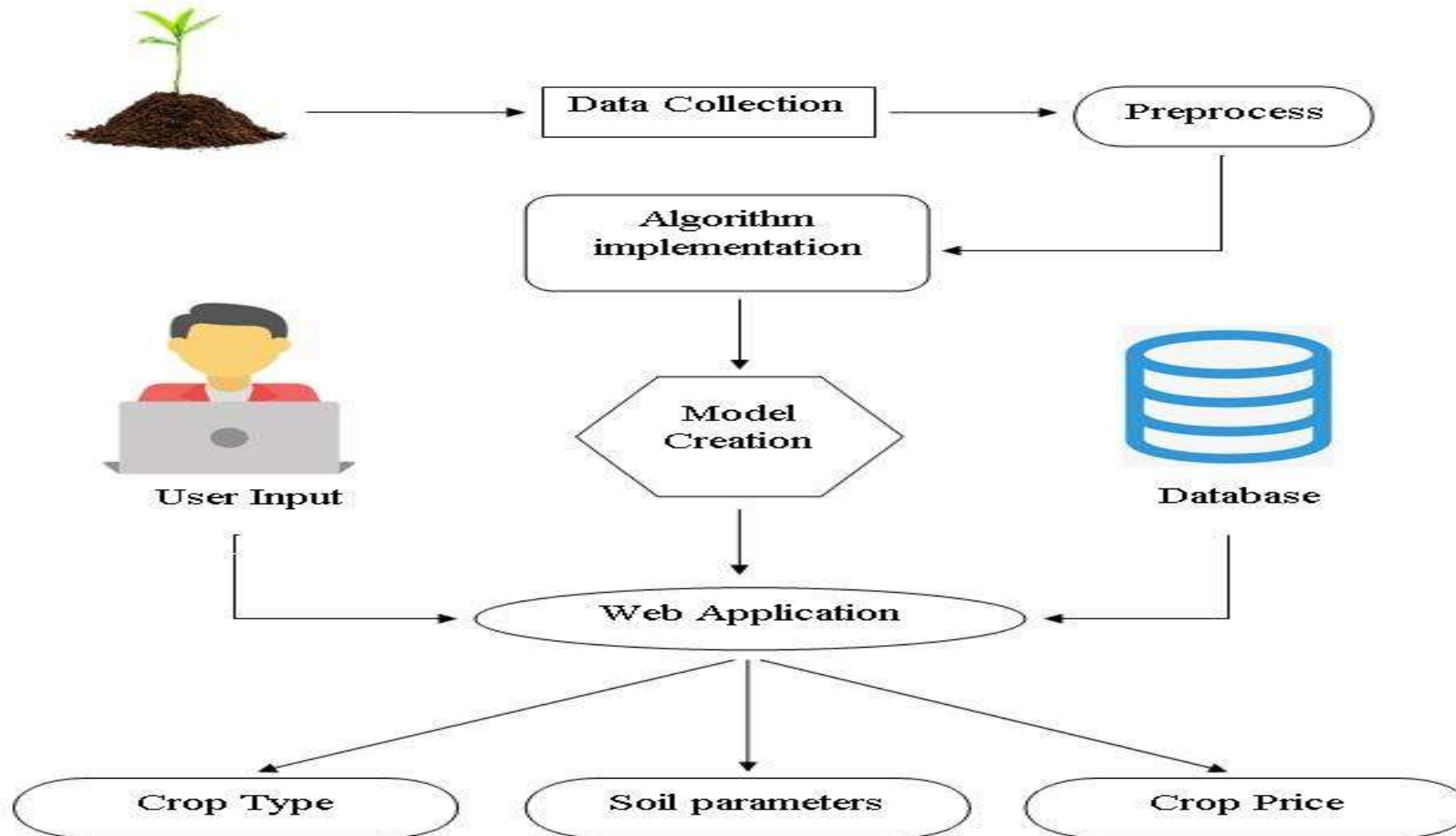


EXISTING SYSTEM

- Traditional farming methods mainly rely on farmers' experience and manual decision-making.
- Limited use of real-time data for soil, weather, and crop condition analysis.
- Late detection of crop diseases and pest attacks, resulting in reduced crop yield.
- Inefficient irrigation and fertilizer management, leading to water and resource wastage.
- Manual record keeping, making farm management time-consuming and less accurate.
- Lack of AI-based decision support, reducing the accuracy of crop selection and farming practices.
- Difficulty in predicting weather changes, which can negatively affect crop growth and harvesting.
- Higher production costs and lower productivity due to inefficient resource management and traditional farming techniques.



ARCHITECTURAL DIAGRAM





PROBLEM STATEMENT

- Traditional farming methods rely on manual decisions, which may lead to inefficient crop management.
- Unpredictable weather conditions make it difficult for farmers to plan cultivation effectively.
- Delayed detection of crop diseases and pest attacks results in reduced crop yield and financial losses.
- Inefficient use of water and fertilizers leads to resource wastage and increased farming costs.
- Lack of AI-based decision support prevents farmers from making accurate, data-driven agricultural decisions.
- Poor management of farm records and agricultural data affects productivity and long-term planning.
- Limited access to accurate agricultural information makes it difficult for farmers to choose the right crops and farming techniques.



PROPOSED SYSTEM

- Develops an AI-based agriculture management system using Java to support smart and sustainable farming.
- Analyzes soil, weather, and crop data to recommend the most suitable crops for cultivation.
- Predicts crop diseases and pest attacks at an early stage using AI, helping farmers reduce crop losses.
- Provides intelligent irrigation and fertilizer recommendations to optimize water usage and improve soil fertility.
- Maintains farmer and crop records in a secure database for efficient farm management.
- Reduces farming costs while increasing crop productivity through efficient resource utilization.
- Promotes sustainable and eco-friendly agricultural practices by minimizing resource wastage and improving overall farm efficiency.



MODULE 1

Priority Queue Scheduling Engine

- **Algorithm:** Priority Queue Scheduling Algorithm (Heap-Based)
- Collects all agricultural tasks generated by the system.
- Assigns execution order based on task priority.
- Ensures high-priority tasks such as disease alerts and irrigation are processed first.
- Dynamically updates the queue whenever new high-priority tasks are added.
- Improves scheduling efficiency and minimizes delays.



MODULE 2

Task Generation and Priority Assignment

- **Algorithm:** Rule-Based Decision Algorithm with Weighted Priority Scoring
- Generates farming tasks from soil, weather, crop, and disease information.
- Calculates a priority score based on predefined rules.
- Classifies tasks into High, Medium, and Low priority.
- Sends prioritized tasks to the Priority Queue.
- Supports intelligent decision-making for farm management.



MODULE 3

Performance Evaluation and Result Analysis

- **Algorithm:** Performance Evaluation Algorithm
- Measures task completion time.
- Calculates average waiting time and scheduling efficiency.
- Evaluates resource utilization.
- Compares Priority Queue scheduling with FCFS scheduling.
- Generates analytical reports for performance improvement.



CONCLUSION



- The AI-Based Sustainable Agriculture Management System improves farming efficiency by using Artificial Intelligence and Java technology.
- The system helps farmers make accurate and timely decisions through crop recommendations, disease prediction, and smart irrigation planning.
- It optimizes the use of water, fertilizers, and other resources, reducing costs and minimizing environmental impact.
- By integrating AI with sustainable farming practices, the system increases crop productivity, enhances farm management, and supports long-term agricultural sustainability.
- The system stores and manages agricultural data efficiently, enabling better planning, monitoring, and decision-making.



FUTURE SCOPE



- Integrate IoT sensors for real-time monitoring of soil moisture, temperature, and humidity.
- Use drones and satellite imagery to monitor crop health and detect diseases over large farming areas.
- Implement advanced AI and Machine Learning models to improve the accuracy of crop recommendations and disease prediction.
- Develop a mobile application to provide farmers with instant access to recommendations and alerts.
- Integrate real-time weather forecasting to support better irrigation and crop planning.
- Support multiple regional languages to make the system accessible to farmers from different regions.
- Enable cloud-based data storage for secure access, backup, and sharing of agricultural records.
- Expand the system with market price prediction to help farmers decide the best time to sell their crops for maximum profit.



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THANK YOU